Quantum Reed-Muller (Steane $[[7,1,3]]$) Code

1. Introduction

Quantum Reed-Muller (RM) codes are a powerful family of CSS codes constructed from classical RM codes or their punctured versions. The Steane $[[7,1,3]]$ code is a prominent punctured RM code (specifically derived from the classical Hamming $[7,4,3]$ code).

Unlike the Shor code, which uses 9 qubits to correct a single error by concatenating repetition codes, the QRM (Steane) code achieves the same distance ($d=3$) using only 7 qubits. Furthermore, because it is a **CSS code**, its X and Z error correction protocols are perfectly symmetrical, and it allows for transversal implementation of certain Clifford gates.

1.1 The Parity Check Matrix

The stabilizer generators are defined by the classical Hamming parity check matrix H:

$$H = \begin{pmatrix} 1 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 \end{pmatrix}$$

Because this matrix is self-orthogonal ($HH^T = 0 \pmod 2$), we can use the exact same matrix for both our Z-stabilizers (to detect X errors) and X-stabilizers (to detect Z errors).

- Each row of H defines a stabilizer generator.
- For Z-checks: Row 1 means we check the parity of qubits 0, 3, 5, and 6 using Z operators.
- For X-checks: Row 1 means we check the parity of the same qubits using X operators.

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2. Qiskit Implementation

This implementation builds the complete 13-qubit system (7 Data + 6 Ancilla) for explicit syndrome extraction. We use 3 ancillas for the Z-checks and 3 ancillas for the X-checks.

In [1]:
```
%pip install numpy==1.26.4 scipy==1.12.0

import numpy as np
import matplotlib.pyplot as plt
from qiskit import QuantumCircuit, QuantumRegister, ClassicalRegister
from qiskit_aer import AerSimulator
from scipy.optimize import fsolve
```

```python
print("Imports successful")
```

```
Requirement already satisfied: numpy==1.26.4 in c:\users\user\anaconda3\lib\site-pac
kages (1.26.4)
Collecting scipy==1.12.0
  Downloading scipy-1.12.0-cp312-cp312-win_amd64.whl.metadata (60 kB)
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Installing collected packages: scipy
  Attempting uninstall: scipy
    Found existing installation: scipy 1.17.1
    Uninstalling scipy-1.17.1:
      Successfully uninstalled scipy-1.17.1
Successfully installed scipy-1.12.0
Note: you may need to restart the kernel to use updated packages.
Imports successful
```

In [2]:

```python
class QuantumReedMuller:
    def __init__(self):
        # 7 Data Qubits
        self.data = QuantumRegister(7, name='data')

        # 6 Ancilla Qubits (3 for Z-checks, 3 for X-checks)
        self.anc = QuantumRegister(6, name='anc')

        # Classical Registers for Syndromes
        self.sz = ClassicalRegister(3, name='sz') # Z-syndromes (detect X flips)
        self.sx = ClassicalRegister(3, name='sx') # X-syndromes (detect Z flips)

        # Final readout
        self.out = ClassicalRegister(7, name='readout')

        self.qc = QuantumCircuit(self.data, self.anc, self.sz, self.sx, self.out)

    def encode_logical_zero(self):
        """
        Encodes |0> into the logical |0>_L state of the Steane code
        Generates a uniform superposition of all codewords in the dual code
        """
        # Create superposition on the message qubits
        self.qc.h(self.data[0:3])

        # Parity generation based on the H matrix
        # q3 = q0 + q1
        self.qc.cx(self.data[0], self.data[3])
        self.qc.cx(self.data[1], self.data[3])
        # q4 = q1 + q2
        self.qc.cx(self.data[1], self.data[4])
        self.qc.cx(self.data[2], self.data[4])
        # q5 = q0 + q2
        self.qc.cx(self.data[0], self.data[5])
        self.qc.cx(self.data[2], self.data[5])
        # q6 = q0 + q1 + q2
        self.qc.cx(self.data[0], self.data[6])
        self.qc.cx(self.data[1], self.data[6])
        self.qc.cx(self.data[2], self.data[6])

        self.qc.barrier(label='Encoded')

    def measure_syndromes(self):
        """
        Non-destructive measurement of the 6 stabilizers
        """
```

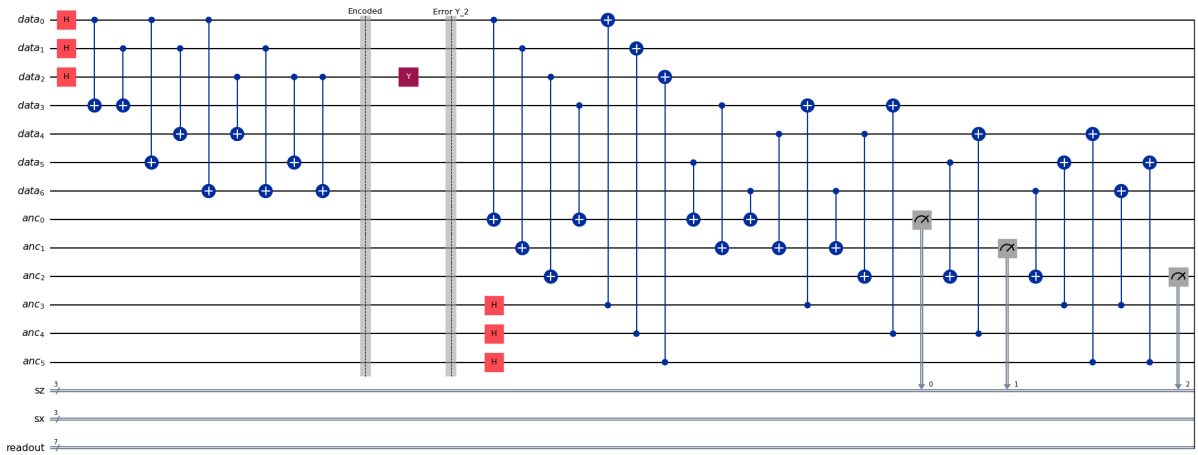
```python
        # Z-Checks (Detect X errors) using Ancillas 0, 1, 2
        # Row 1 - Qubits 0, 3, 5, 6
        self.qc.cx(self.data[0], self.anc[0]); self.qc.cx(self.data[3], self.anc[0]
        self.qc.cx(self.data[5], self.anc[0]); self.qc.cx(self.data[6], self.anc[0]

        # Row 2 - Qubits 1, 3, 4, 6
        self.qc.cx(self.data[1], self.anc[1]); self.qc.cx(self.data[3], self.anc[1]
        self.qc.cx(self.data[4], self.anc[1]); self.qc.cx(self.data[6], self.anc[1]

        # Row 3 - Qubits 2, 4, 5, 6
        self.qc.cx(self.data[2], self.anc[2]); self.qc.cx(self.data[4], self.anc[2]
        self.qc.cx(self.data[5], self.anc[2]); self.qc.cx(self.data[6], self.anc[2]

        # X-Checks (Detect Z errors) using Ancillas 3, 4, 5
        self.qc.h(self.anc[3:6]) # Basis change for X measurements

        # Row 1
        self.qc.cx(self.anc[3], self.data[0]); self.qc.cx(self.anc[3], self.data[3]
        self.qc.cx(self.anc[3], self.data[5]); self.qc.cx(self.anc[3], self.data[6]
        # Row 2
        self.qc.cx(self.anc[4], self.data[1]); self.qc.cx(self.anc[4], self.data[3]
        self.qc.cx(self.anc[4], self.data[4]); self.qc.cx(self.anc[4], self.data[6]
        # Row 3
        self.qc.cx(self.anc[5], self.data[2]); self.qc.cx(self.anc[5], self.data[4]
        self.qc.cx(self.anc[5], self.data[5]); self.qc.cx(self.anc[5], self.data[6]

        self.qc.h(self.anc[3:6])

        # Measurements
        self.qc.measure(self.anc[0:3], self.sz)
        self.qc.measure(self.anc[3:6], self.sx)
        self.qc.barrier(label='Syndrome Extracted')

    def apply_correction(self):
        """
        Applies dynamic correction
        The binary value of the syndrome points directly to the index of
        the faulty qubit
        """
        # Dictionary mapping integer syndrome to qubit index
        # Synergy of the Hamming matrix:
        # syn=1 (001)->q0, syn=2 (010)->q1, syn=4 (100)->q2
        # syn=3 (011)->q3, syn=6 (110)->q4, syn=5 (101)->q5, syn=7 (111)->q6

        # Correct X Errors (Bit flips) using Z-syndrome
        with self.qc.if_test((self.sz, 1)): self.qc.x(self.data[0])
        with self.qc.if_test((self.sz, 2)): self.qc.x(self.data[1])
        with self.qc.if_test((self.sz, 4)): self.qc.x(self.data[2])
        with self.qc.if_test((self.sz, 3)): self.qc.x(self.data[3])
        with self.qc.if_test((self.sz, 6)): self.qc.x(self.data[4])
        with self.qc.if_test((self.sz, 5)): self.qc.x(self.data[5])
        with self.qc.if_test((self.sz, 7)): self.qc.x(self.data[6])

        # Correct Z Errors (Phase flips) using X-syndrome
        with self.qc.if_test((self.sx, 1)): self.qc.z(self.data[0])
        with self.qc.if_test((self.sx, 2)): self.qc.z(self.data[1])
```

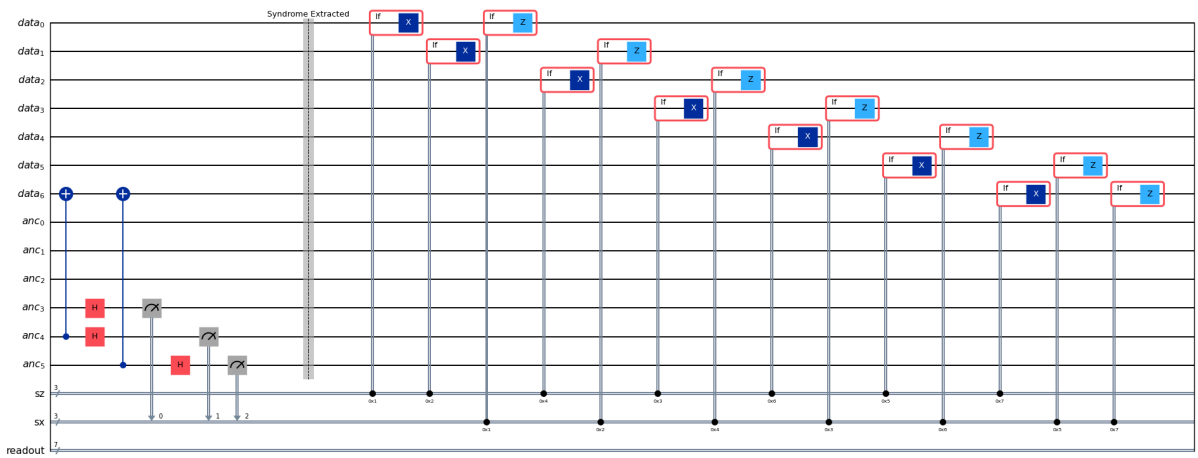
```
        with self.qc.if_test((self.sx, 4)): self.qc.z(self.data[2])
        with self.qc.if_test((self.sx, 3)): self.qc.z(self.data[3])
        with self.qc.if_test((self.sx, 6)): self.qc.z(self.data[4])
        with self.qc.if_test((self.sx, 5)): self.qc.z(self.data[5])
        with self.qc.if_test((self.sx, 7)): self.qc.z(self.data[6])

        self.qc.barrier(label='Corrected')

    def decode_and_verify(self):
        """
        Uncomputes the encoding sequence. If correct, returns to |0000000>
        """
        self.qc.cx(self.data[0], self.data[6]); self.qc.cx(self.data[1], self.data[
        self.qc.cx(self.data[0], self.data[5]); self.qc.cx(self.data[2], self.data[
        self.qc.cx(self.data[1], self.data[4]); self.qc.cx(self.data[2], self.data[
        self.qc.cx(self.data[0], self.data[3]); self.qc.cx(self.data[1], self.data[

        self.qc.h(self.data[0:3])
        self.qc.measure(self.data, self.out)
        return self.qc
```

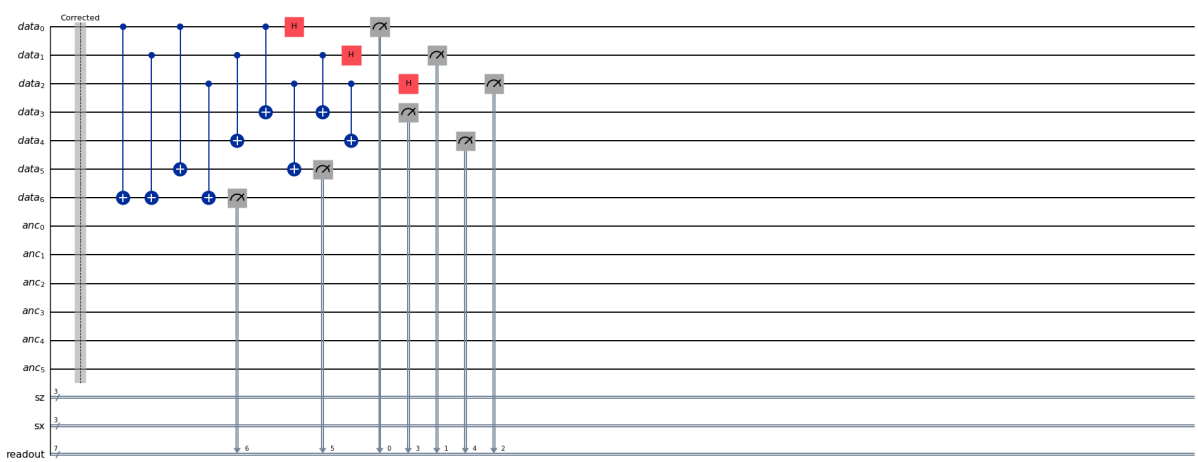
In [3]:

```
qrm_demo = QuantumReedMuller()
qrm_demo.encode_logical_zero()

# Inject a manual error for demonstration (e.g., Y error on Qubit 2)
print("Injecting Y error on Qubit 2...")
qrm_demo.qc.y(qrm_demo.data[2])
qrm_demo.qc.barrier(label='Error Y_2')

# Complete the cycle
qrm_demo.measure_syndromes()
qrm_demo.apply_correction()
qrm_circ = qrm_demo.decode_and_verify()

# Draw the circuit
# fold=-1 prevents the circuit from wrapping to a new line
# scale=0.7 shrinks the 13-qubit height to fit screen
display(qrm_circ.draw('mpl', fold=40, scale=0.7))
```

```
Injecting Y error on Qubit 2...
```







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3. Performance Simulation

To evaluate the performance of the Steane $[[7, 1, 3]]$ code, we inject independent depolarizing noise (X, Y, or Z) on all 7 qubits.

Because the CSS construction separates X and Z tracking, the code can occasionally correct weight-2 errors if they occur in different bases (for example, an X error on q_0 and a Z error

on q_1). However, the theoretical worst-case bound assumes the code fails if *any* 2 or more errors occur across the 7 qubits.

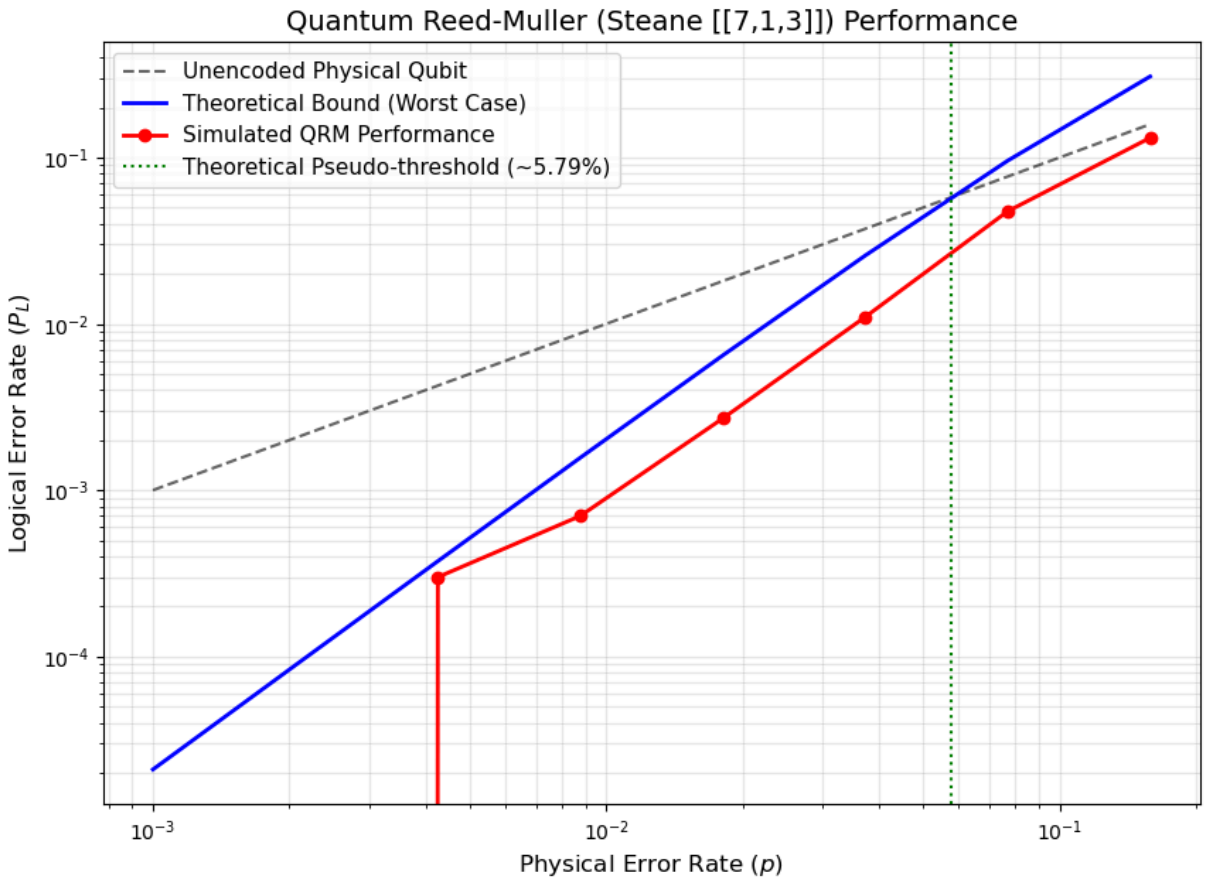
We will compare our simulated logical error rate against the theoretical worst-case failure probability:

$$P_{theory} = 1 - (1-p)^7 - 7p(1-p)^6$$

In [ ]:
```
def apply_random_noise(qrm_instance, p):
    for qubit_idx in range(7):
        if np.random.random() < p:
            error_type = np.random.choice(['X', 'Y', 'Z'])
            if error_type == 'X': qrm_instance.qc.x(qrm_instance.data[qubit_idx])
            elif error_type == 'Y': qrm_instance.qc.y(qrm_instance.data[qubit_idx])
            elif error_type == 'Z': qrm_instance.qc.z(qrm_instance.data[qubit_idx])

def run_qrm_simulation(p_values, shots_per_point=1000):
    logical_error_rates = []
    # Stabilizer is fast and handles CSS
    backend = AerSimulator(method='stabilizer')

    print(f"Simulating {len(p_values)} points...")
    for p in p_values:
        failures = 0
        circuits = []

        for _ in range(shots_per_point):
            qrm = QuantumReedMuller()
            qrm.encode_logical_zero()
            apply_random_noise(qrm, p)
            qrm.measure_syndromes()
            qrm.apply_correction()
            circuits.append(qrm.decode_and_verify())

        result = backend.run(circuits, shots=1, optimization_level=0).result()

        for i in range(shots_per_point):
            counts = result.get_counts(i)
            outcome = list(counts.keys())[0]
            readout = outcome.split(" ")[0]
            if readout != '0000000':
                failures += 1

        p_logical = failures / shots_per_point
        logical_error_rates.append(p_logical)
        print(f"p={p:.4f} -> P_Logical={p_logical:.4f}")

    return logical_error_rates

# Define physical error rates
p_phys_samples = np.logspace(-3, -0.8, 8)

# Run simulation
p_log_simulated = run_qrm_simulation(p_phys_samples, shots_per_point=10000)
```

```python
# Calculate Theoretical Bounds
p_theoretical = 1 - (1 - p_phys_samples)**7 - 7 * p_phys_samples * (1 - p_phys_samp

# Dynamically calculate the exact theoretical crossover
def theoretical_diff(p):
    """The difference between the theoretical bound and the unencoded baseline"""
    return (1 - (1 - p)**7 - 7 * p * (1 - p)**6) - p

# fsolve finds where the difference is 0. We give it an initial guess of 0.05
exact_crossover = fsolve(theoretical_diff, 0.05)[0]

# Plotting
plt.figure(figsize=(10, 7))

plt.loglog(p_phys_samples, p_phys_samples, 'k--', label='Unencoded Physical Qubit',
plt.loglog(p_phys_samples, p_theoretical, 'b-', label='Theoretical Bound (Worst Cas
plt.loglog(p_phys_samples, p_log_simulated, 'ro-', label='Simulated QRM Performance

# Use the exact calculated value for the vertical line
plt.axvline(x=exact_crossover, color='green', linestyle=':',
            label=f'Theoretical Pseudo-threshold (~{exact_crossover*100:.2f}%)')

plt.title('Quantum Reed-Muller (Steane [[7,1,3]]) Performance', fontsize=14)
plt.xlabel('Physical Error Rate ($p$)', fontsize=12)
plt.ylabel('Logical Error Rate ($P_L$)', fontsize=12)
plt.legend(fontsize=11)
plt.grid(True, which="both", ls="-", alpha=0.3)

plt.show()
```

```
Simulating 8 points...
p=0.0010 -> P_Logical=0.0000
p=0.0021 -> P_Logical=0.0000
p=0.0043 -> P_Logical=0.0003
p=0.0088 -> P_Logical=0.0007
p=0.0181 -> P_Logical=0.0027
p=0.0373 -> P_Logical=0.0110
p=0.0769 -> P_Logical=0.0475
p=0.1585 -> P_Logical=0.1314
```


Quantum Reed-Muller (Steane [[7,1,3]]) Performance
Unencoded Physical Qubit
Theoretical Bound (Worst Case)
Simulated QRM Performance
Theoretical Pseudo-threshold (~5.79%)
Logical Error Rate (P_L)
Physical Error Rate (p)
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-3}
10^{-2}
10^{-1}